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SHIELDING DESIGN OF THE ITER NBI DUCT FOR NUCLEAR AND BREMSSTRAHLUNG RADIATION

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Three-dimensional Monte Carlo shielding analyses are conducted on the ITER Neutral Beam Injection (NBI) duct for the nuclear and bremsstrahlung radiation. The detailed distribution is evaluated about the nuclear heating rate and surface heat load of the NBI duct wall by the neutron and photon transport calculation. The analytical representations of these nuclear responses are established as a function of the distance from the blanket surface. It is clarified that these representations are different between the duct wall facing the plasma and that hidden from the plasma, and also between the duct wall in the blanket region and that in the vacuum vessel region. These results are very useful for the shielding design of the NBI duct wall in the nuclear fusion reactor.

INTRODUCTION

In the ITER, many ducts penetrating the blanket and vacuum vessel are located in the confined space between adjacent super-conductive Toroidal Field (TF) coils. In order to reduce the radiation streaming through these ducts, the shield plug will be installed inside the duct. However, the shield plug cannot be installed inside the Neutral Beam Injection (NBI) duct in order to inject the neutral beam to the plasma directly through the duct. The nuclear heating rate and surface heat load in the NBI duct wall are enhanced by the radiation streaming, so these are a critical concern in the shielding design.

In the previous studies, the detailed evaluation has not been reported about the nuclear heating rate and surface heat load in the NBI duct wall, although many results have been reported about the shielding calculation of the TF coil and the decay gamma-ray dose rates around the NBI duct[1–6]. In the present study, three-dimensional (3-D) Monte Carlo calculations are conducted with calculation geometries simulating the overall region of the tokamak, and the nuclear heating rate and surface heat load in the duct wall are evaluated for the source distributions owing to the nuclear fusion reaction and bremsstrahlung radiation from the plasma, respectively.

CALCULATION METHODS

Nuclear heating rate

The neutron and photon transport calculations were conducted by the Monte Carlo neutral particle transport code MCNP-4C[7] with the Fusion Evaluated Nuclear Data Library (FENDL)[8]. The detail of the calculation model is shown in Ref. (6). The horizontal cross-sectional view is shown in Figure 1 about the calculation model around the NBI duct. The duct opening inlet is ~58 cm in width and 123 cm in height. The blanket is ~45 cm in thickness. The NBI duct wall in the right-hand side (facing surface) faces the much wider plasma region compared with that in the left-hand side (hidden surface). There is a ~5 cm long gap between the duct wall in the blanket region and that in the vacuum vessel. The nuclear fusion neutron source distribution is given by the plasma equilibrium calculation[9], and it is generated under the condition of the total nuclear fusion power being 500 MW.

The cell volume is required to be estimated to evaluate the nuclear heating rate per unit volume. The configuration of the cell is very complex in this calculation model, so the cell volume cannot be analytically calculated. The cell volume can be estimated by the Monte Carlo calculation in this study. The defined materials of the cell are tentatively replaced to the void in all cells in this process. The parallel neutron surface source fully covering the estimated cell region is generated, and the Monte Carlo calculation is conducted. The weight of the generated source is defined to be area of the surface source. All boundaries are defined as the vacuum boundary to omit the return of the neutron at the boundary. The track length is counted by each cell by the Monte Carlo calculation. The track length in each cell can be considered the volume.

Surface heat load

The photon transport calculations were conducted using MCNP-4C and FENDL with the same model used in the evaluation of the nuclear heating rate. The photon with the energy of 10 keV is generated according to the bremsstrahlung and line radiation source distributions, which are given by the plasma equilibrium calculation[9], under the condition of the total radiation power being 136 MW, and the Monte

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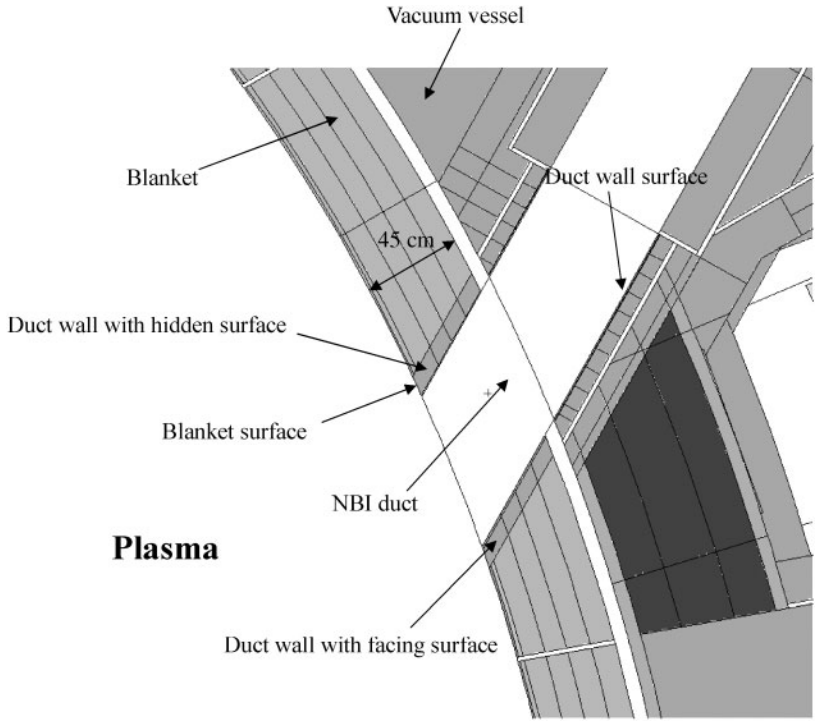


Figure 1. Horizontal cross-sectional view of the calculation model around the NBI duct.

Carlo calculation is conducted. The thickness of the cell in the duct wall surface is defined to be very thin to neglect the incident radiation through the surface except for the duct wall surface, and it is 0.05 mm. All energy of the photon entering the cell can be deposited in the cell with 0.05 mm thickness. The energy deposition is calculated by each cell, and the surface heat load is evaluated.

RESULTS AND DISCUSSIONS

Nuclear heating rate

Figures 2 and 3 show the nuclear heating rate in the duct walls with the facing and hidden surfaces, respectively, at the locations 2 and 6 cm from the duct wall surface. The distance shown in the horizontal axis means the length from the blanket surface to the centre of the cell along the duct wall surface. The nuclear heating rates are almost the same between the both walls at the blanket surface, and these are 5–7 W cm^{-3}. The analytical representation on the nuclear heating rate distribution, Y (W cm^{-3}), can be given as a function of the distance from the blanket surface, X (cm), as follows:

$$Y = C \exp(-\lambda X). \quad (1)$$

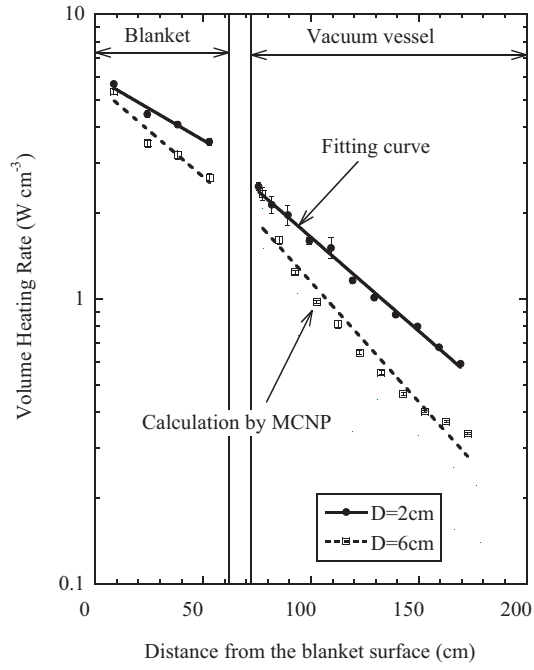


Figure 2. Nuclear heating rate in the duct walls with the facing surface at the locations 2 and 6 cm from the duct wall surface.

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The values of C and λ obtained by the least squares fitting to Equation 1 are shown in Table 1 for the locations 0.5, 2, 6, 10 and 14 cm from the duct wall surface.

The values of λ of the facing surface is much smaller than those of the hidden surface in the blanket region, while most of these values are almost similar between the facing and hidden surfaces at the locations of same distance from the duct wall surface in the vacuum vessel region. In the facing surface, these values in the blanket region is smaller than that in the vacuum vessel region. In the hidden surface, the former is larger than the latter by a factor of more than 2. Most of these values increase, as the distance is longer from the duct wall surface. These values are given by the following formula

$$\lambda_{F,B} < \lambda_{F,V} \approx \lambda_{H,V} < \lambda_{H,B}, \quad (2)$$

where $\lambda_{F,B}$, $\lambda_{F,V}$, $\lambda_{H,V}$ and $\lambda_{H,B}$ are the values of the facing surface in the blanket region, that in the vacuum vessel region, that of the hidden surface in the vacuum vessel region and that in the blanket region, respectively. These values depend on the ratio of the incident neutron flux through the duct to that through the blanket surface. When the ratio is larger, the value becomes smaller. The attenuation of the neutron current through the blanket surface is much larger than that through the duct due to the shielding of the blanket itself. The solid angles, whose the duct wall can look the plasma directly, in the facing surface are much larger than those in the hidden surface, so the ratio of the facing surface is larger than that of the hidden surface. As a result, the nuclear heating rates in the vacuum vessel region in the facing surface are larger than those in the hidden surface by a factor of ~3 at the location of 2 cm from the duct wall surface.

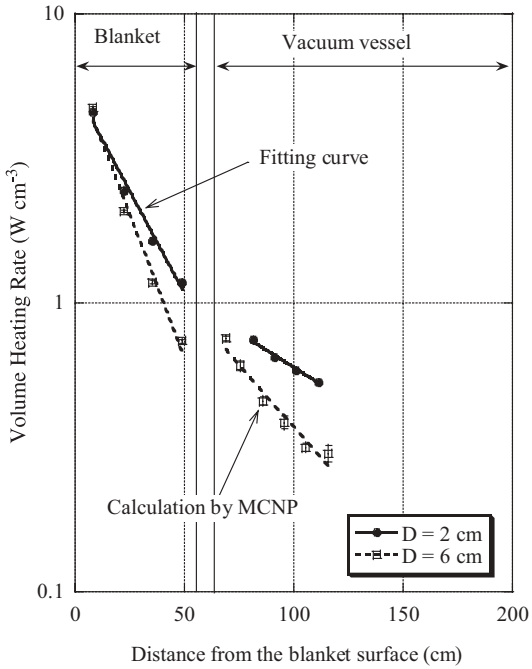


Figure 3. Nuclear heating rate in the duct walls with the hidden surface at the locations 2 and 6 cm from the duct wall surface.

Surface heat load

Figure 4 shows the surface heat load as a function of the distance from the blanket surface. The surface heat load is 17–18 W cm^{-2} at the blanket surface in the facing surface. The surface heat loads in the hidden surfaces are 4–7 times lower than those in the facing surfaces in the blanket region, and the former are 3–5 times lower than the latter in the vacuum vessel region. The surface heat load is ~2–3 W cm^{-2} at the blanket surface in the hidden surface. The surface heat loads in the facing and hidden surfaces are 2–3 and 0.6–0.7 W cm^{-2}, respectively, at the location 1 m from the blanket surface. The surface heat load is generated by only the non-collision radiation through the duct from

Table 1. Values of C and λ obtained by the least squares fitting to Equation 1, nuclear heating rate as a function of the distance from the blanket surface, for the locations 0.5, 2, 6, 10 and 14 cm from the duct wall surface.

Distance from the duct wall surface (cm)	Facing surface				Hidden surface			
	Blanket region		Vacuum vessel region		Blanket region		Vacuum vessel region	
	C	λ	C	λ	C	λ	C	λ
0.5	5.8	0.0076	7.6	0.014	5.3	0.025	2.7	0.011
2	6.0	0.010	7.5	0.015	5.5	0.033	3.0	0.015
6	5.6	0.015	8.0	0.019	6.3	0.045	2.7	0.020
10	5.3	0.020	9.1	0.022	6.8	0.058	3.8	0.030
14	4.7	0.023	6.9	0.023	7.8	0.066	10.8	0.049

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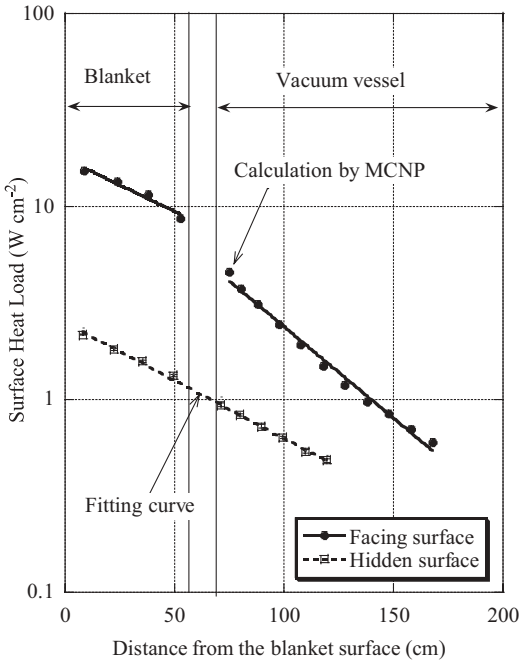


Figure 4. Surface heat load in the duct walls with the facing and hidden surfaces as a function of the distance from the blanket surface.

the plasma. The intensity of the non-collision radiation entering the duct wall depends on the solid angle looking the plasma directly. As a result, the surface heat load in the facing surface is much larger than that in the hidden surface, and the difference between both about the surface heat load is larger than that about the nuclear heating rate.

Similar to the nuclear heating rate, the surface heat load distribution can be given as a function of the distance from the blanket surface (Equation 1). In the facing surface, the value of λ in the vacuum vessel region drastically increases compared with that in the blanket region. Because the solid angles drastically decrease at the boundary between the blanket region and the vacuum vessel. The values of C and λ in Equation 1 are 17.7 and 0.013, respectively, in the blanket region in the facing surface, and they are 21.5 and 0.022, respectively, in the vacuum vessel region. The value of λ is almost constant in the hidden surface, and the values of C and λ in Equation 1 are 2.5 and 0.014, respectively.

CONCLUSION

The detailed distribution of the nuclear heating rate and surface heat load of the ITER NBI duct wall was evaluated by the 3-D Monte Carlo neutron and photon transport calculation. The analytical representations of these nuclear responses were established as a function of the distance from the blanket surface. It was clarified that these are different between the duct wall facing the plasma and that hidden from the plasma, and also between the duct wall in the blanket region and that in the vacuum vessel region. The nuclear heating rates in the vacuum vessel region in the facing surface are larger than those in the hidden surface by a factor of about three at the location of 2 cm distance from the duct wall surface. The surface heat loads in the facing surface are 4–7 times higher than those in the hidden surface in the blanket region, and the former are 3–5 times higher than the latter in the vacuum vessel region. These results are very useful for the shielding design of the NBI duct wall in the nuclear fusion reactor.

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